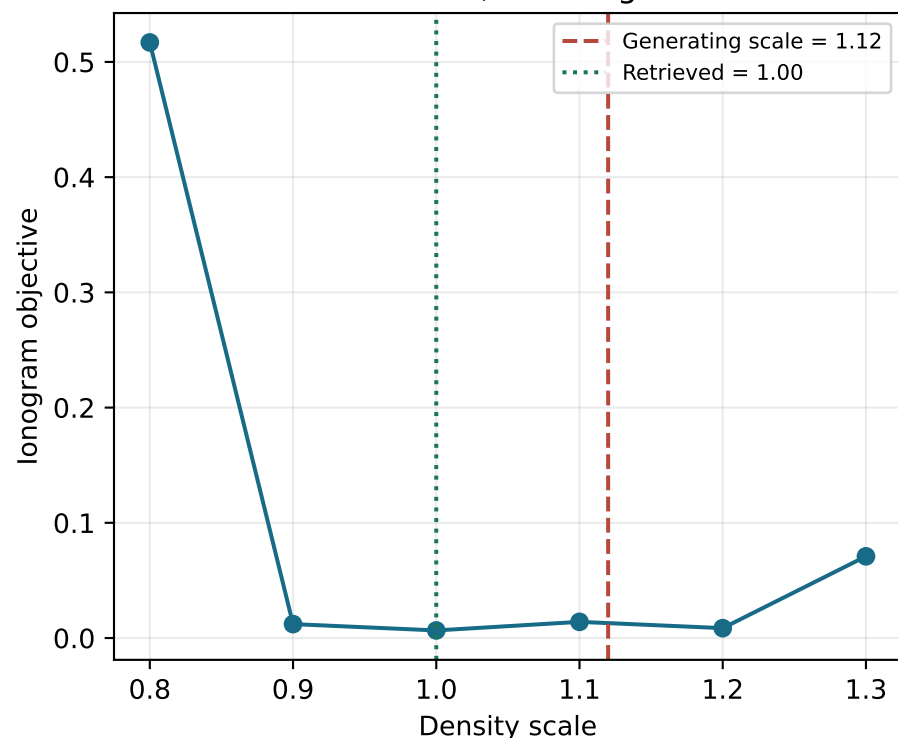
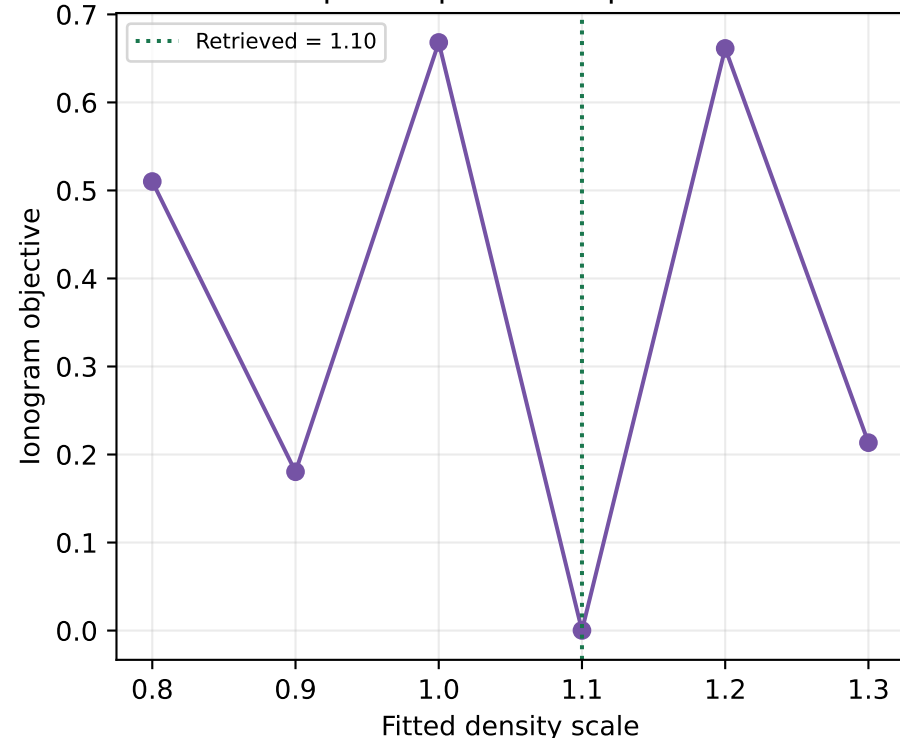


Topside retrieval diagnostic — generated Earth case, not Mars validation

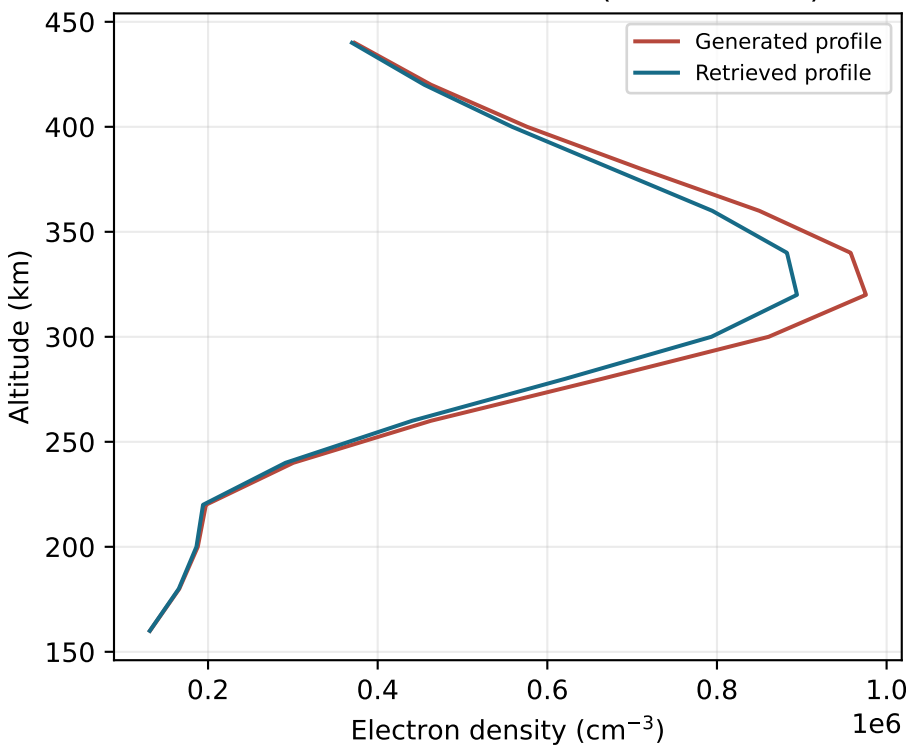
A Closed-model, coarse grid search



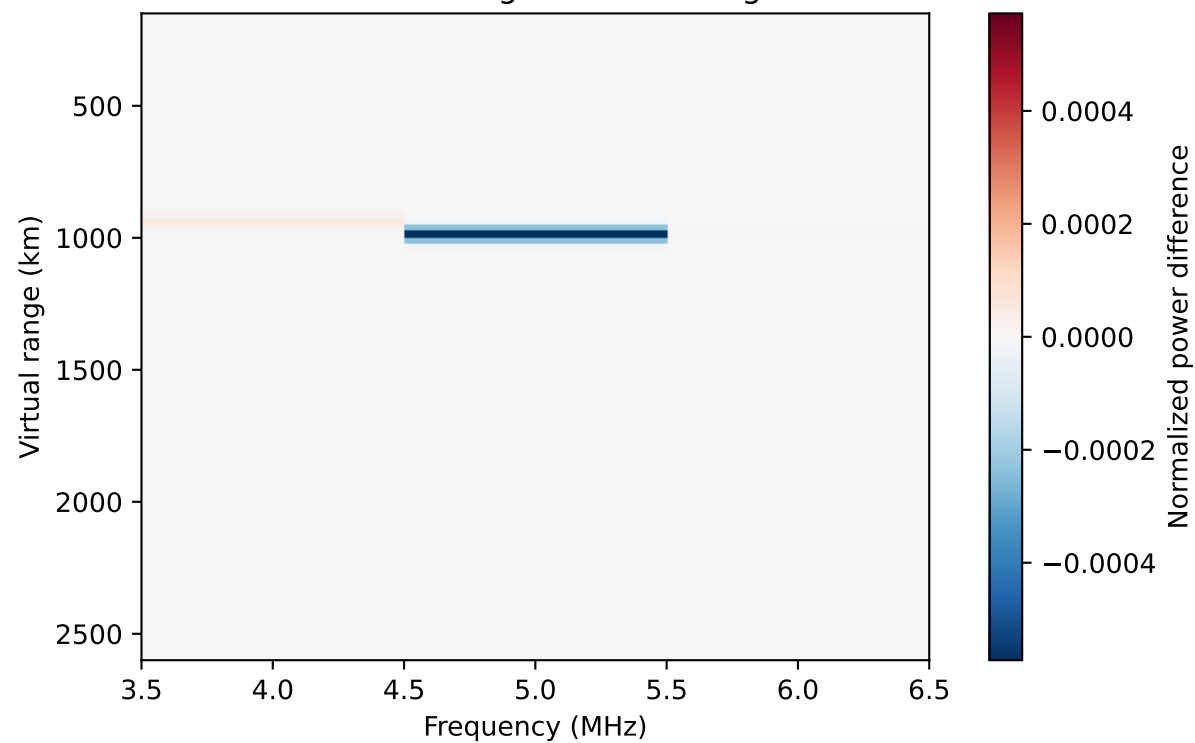
B Separate profile-shape formula



C Profile NRMSE = 6.7% (150-450 km)



D Fitted minus generated ionogram



Topside retrieval: evidence and status

25 September 2026 | raytracing / python_raytrace

What the figures establish

Closed-model test: true density scale 1.12; six-point grid search returned 1.00; absolute scale error 0.12; objective 0.0065903. The true value is off-grid.

Separate profile-shape test: best scale 1.10; 150–450 km profile NRMSE 6.7%; ionogram objective 0.00014834. This tests model mismatch, not a Mars measurement.

No-return control: both truth and a scale-0.80 candidate have zero ionogram power, giving objective zero despite 32.0% profile NRMSE. A zero objective alone is therefore not an accuracy claim.

Truth independence audit

The existing demo generates truth with `simulate_dataset` and fits it with `simulate_dataset` using the same PyIRI background, parameterized density field, and PHaRLAP ray tracer. Its truth is therefore not independent of retrieval assumptions.

The new structural check generates the density profile with a separate altitude formula, outside the fitted scalar family. It still shares PyIRI, ray tracing, image formation, and generated Earth orbit geometry. It is only partially independent.

No independent empirical Mars truth or measured Mars ionogram has yet been run through the retrieval. Accordingly, this report does not claim real-world retrieval accuracy.

Public Mars reference candidate

Huang et al. (2021) publish Mathematica expressions for 12 dayside ion species over 150–450 km, driven by EUV index, solar zenith angle, and magnetic elevation. Charge-balanced total ion density is a possible external electron-density reference, pending unit and license checks.

github.com/hamsterping/Empirical-models-of-ion-density-distribution-in-the-dayside-Martian-ionosphere

doi.org/10.1029/2021JA029226

NeMars is a closer MARSIS electron-density model (doi.org/10.1016/j.icarus.2013.03.021), but I did not locate runnable public code. No public Mars-IRI or Mars-NeQuick implementation was verified.

Next validation gate

Drive observations from an external Mars density grid or measured MARSIS profiles, with Mars geometry and magnetic/neutral assumptions, then compare retrieved density to withheld truth on a common altitude and location grid. Report bias, NRMSE, and uncertainty across multiple cases.